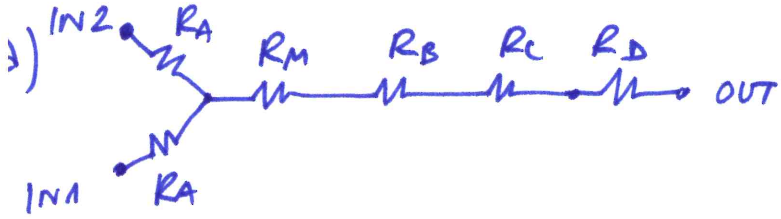
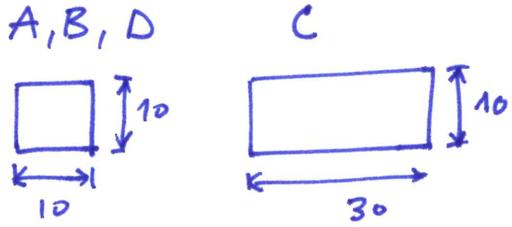
EXERCISE 1

a)



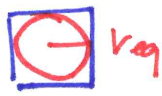


$R = f\left(\frac{1}{w}\right)\cdot L$ to maximize L for fixed $R = 10^{16}\left[\frac{Pa\cdot s}{m^3}\right]$

↳ MAX $w = 20\mu m$

$$R = \frac{12\cdot 10^{-3}}{\left(1-0.63\frac{1}{2}\right)(10\mu m)^3\cdot 20\mu m}\cdot L|_{max} = \boxed{11.4\ mm}$$

$$R_C = \frac{12\cdot 10^{-3}}{\left(1-0.63\frac{1}{3}\right)(10\mu m)^3\cdot 30\mu m}\cdot 2mm = 10^{15}$$



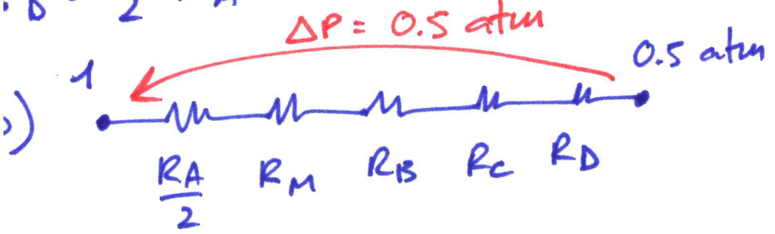
$$R_A = \frac{8\cdot\eta}{\pi\cdot r_{eq}^4}\cdot A = \frac{8\cdot 10^{-3}}{\pi(5\mu m)^4}\cdot 1mm = 4\cdot 10^{15}$$

$r_{eq} = 5\mu m$

$R_B = R_A$

$R_D = \frac{1}{2}R_A = 2\cdot 10^{15}$

b)



$$R_{TOT} = 2\cdot 10^{15} + 10^{16} + 4\cdot 10^{15} + 10^{15} + 2\cdot 10^{15} = 1.9\cdot 10^{16}\left[\frac{Pa\cdot s}{m^3}\right]$$

$$Q = \frac{0.5\cdot 101\ kPa}{1.9\cdot 10^{16}\ Pa\cdot s}\, m^3 = 2.66\cdot 10^{-12}\frac{m^3}{s} = 0.0027\frac{\mu l}{s}\ \left(0{,}16\frac{\mu l}{min}\right)$$

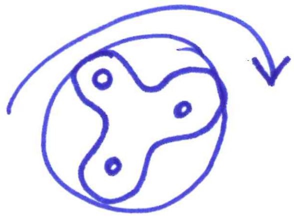
$$v_A = \frac{Q}{2}\cdot\frac{1}{h\cdot w_A} = \boxed{13.3\ mm/s}$$

$$v_D = \frac{Q}{h\cdot w_D} = v_b$$

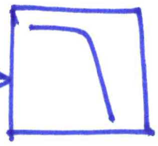
$$v_b = \frac{Q}{h\cdot w_b} = \boxed{26.6\ mm/s}$$

$$\boxed{v_{MIXER} = \frac{Q}{h\cdot 20\mu} = 13.3\ mm/s}$$

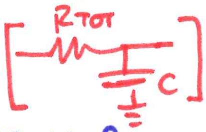
$$v_C = \frac{Q}{h\cdot w_C} = \boxed{8.9\ mm/s}$$

c) 

$$\frac{200\ \text{rpm} \cdot 3}{60} = 10\ \text{Hz}$$ RIPPLE FREQUENCY

SOLUTION → PUT a LPF →  at 1 Hz or below

$$\tau = \frac{1}{2\pi \cdot 1} = 160\ \text{ms} = R \cdot C$$ ← INTRODUCE A FLUIDIC CAPACITANCE



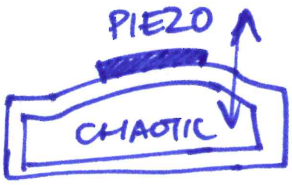
$$C = \frac{\tau}{R} = \frac{160\ \text{ms}}{1.9 \cdot 10^{16}\ \frac{Pa \cdot s}{m^3}} = \boxed{8.37 \cdot 10^{18}} \left[\frac{m^3}{Pa}\right]$$

d) T < Diffusion $D \propto T$

Viscosity $\eta \propto \frac{1}{T}$ weak dependence

$T\uparrow \Rightarrow \eta\downarrow \Rightarrow R\downarrow \rightarrow v\uparrow\ t\downarrow$ $\quad L \sim \sqrt{D \cdot t} \sim \sqrt{\underbrace{D\uparrow}_{\text{DOMINATES}} \cdot t\downarrow}$

DOMINATES ⇓ MIXING IMPROVES

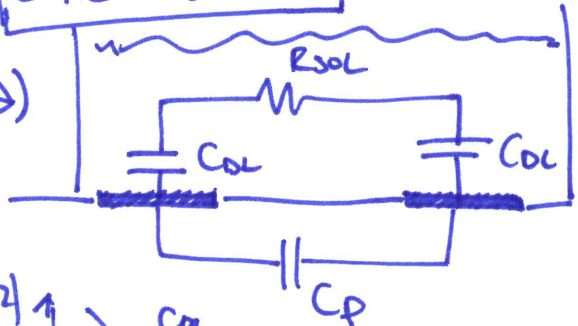


e) ADDITIVE: SERIAL, NO NEED FOR MOLDS, @ LAB WORSE RESOLUTION

REPLICA MOLDING: PARALLELISABLE, NEED MASK / MOLD, BETTER RESOLUTION

EXERCISE 2

a)

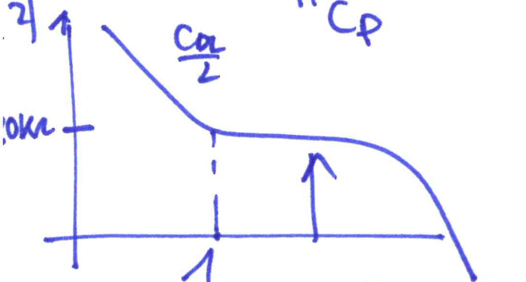


$$C_{DL} = AREA \cdot 0.1 \frac{pF}{\mu m^2}$$

$$= \underset{2}{S} \cdot \underset{50}{E \cdot F} \cdot 0.1 \frac{pF}{\mu m^2} = 50\,pF$$

$$C_P = C_{water} + C_{stray}$$

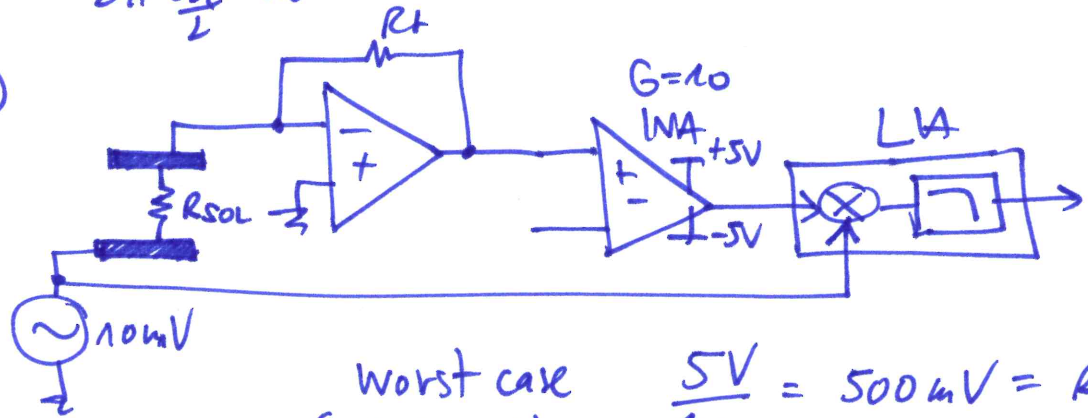
b)



$$\frac{1}{2\pi \frac{C_{DL}}{2} \cdot R_{sol}} = 32\,\text{kHz}$$

$$f_{sense} > 32\,\text{kHz}$$

c)



Worst case (unbalanced) $\frac{5V}{10} = 500\,mV = R_f \cdot I_{max}$, $I_{max} = \frac{10\,mV}{R_{sol}}$

$$R_f = \frac{500\,mV}{10\,mV} R_{sol} = 50 \cdot 20\,k\Omega = 1\,M\Omega$$

Check BW: $\frac{1}{2\pi 1M\Omega \cdot 0.2pF} = 80\,\text{kHz}$ (tight) → might reduce R_f

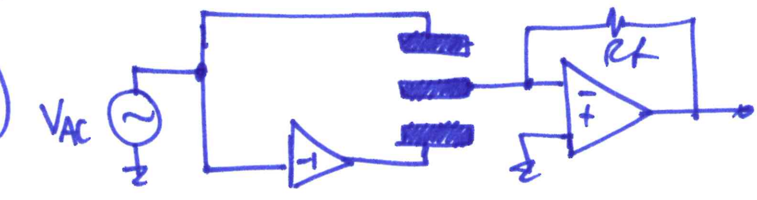
d)

DOMINANT R_{sol} ($R_f \gg R_{sol}$), neglect $\frac{1}{2}$ factor ($\& \frac{\pi}{2}$ ENB)

$$\sqrt{\frac{4kT}{R_{sol}} \cdot BW} = 3pA \Rightarrow BW = \frac{(3pA)^2 \cdot R_{sol}}{4kT} = 10.8\,\text{Hz}$$

e) CHANGE RECEPTOR (ANTIGEN)

f)



CURRENT DIFFERENCE